

Khandaker Rafiul Islam

Full-Stack Developer · CSE Undergraduate

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SUMMARY

Computer Science undergraduate and full-stack developer who builds and ships products end to end, with a particular interest in backend systems. Most recently designed and deployed Oi Tesla Pool, a ride-pooling platform whose booking logic stays correct under concurrent requests through atomic SQL, row-level locking and a finite-state machine. Comfortable across the stack, from PostgreSQL schema design and REST APIs to responsive React interfaces, and driven by the unglamorous details that keep a system correct in production.

TECHNICAL SKILLS

Languages	Go, Python, C++, TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3
Frameworks	Node.js, Express.js, React, Next.js, React Native, FastAPI, Gin (Go), Tailwind CSS
Data & Backend	PostgreSQL, MySQL, MongoDB, Redis, Supabase, Firebase, RESTful APIs, JWT, Microservices
DevOps & Tools	Docker, Docker Compose, Nginx, Prometheus, Grafana, Git, GitHub Actions, CI/CD, Linux/Bash, Postman, Vercel, Render, Jest, PyTest
Core CS	Data Structures & Algorithms, System Design, Operating Systems, Database Systems

TECHNICAL PROJECTS

Oi Tesla Pool | Direction-Aware Ride Pooling

[🔗 Live](#) | [📄 Code](#)

Full-Stack & Backend Developer — React, TypeScript, Node.js/Express, PostgreSQL, Supabase

Sep 2026

- Built a ride-pooling platform for Dhaka's battery rickshaws: riders heading within 90° of each other share one vehicle and each pay 20% less, gated by same-pickup-zone and 3 km detour-cap rules.
- Made seat booking race-safe with a single atomic conditional UPDATE and PostgreSQL row locks (SELECT ... FOR UPDATE), so concurrent riders can never double-book the last seat.
- Modelled every ride and pool lifecycle as a finite-state machine and enforced invariants in the database with partial unique indexes (one active ride per rider, one active pool per vehicle).
- Shipped rider and driver features: a shared pool countdown (halvable once per booking), 15-minute request expiry with bonuses, gender-only co-rider visibility for privacy, live ride tracking, ratings, and a driver console.
- Computed today and all-time driver earnings in one query (SUM ... FILTER) backed by a partial index; stored money as integer paisa and validated every request with zod, returning DTOs only.
- Integrated Supabase Auth (email, Google and LinkedIn OAuth) with a DB trigger provisioning profiles; deployed the React/Vite frontend on Vercel, the Express API on Render and PostgreSQL on Supabase.

GravityCloud | Self-Scaling Distributed AI Infrastructure

[📄 Code](#)

Backend & DevOps Developer — Python, FastAPI, Docker, Nginx, Redis, Ollama, ChromaDB, Prometheus, Grafana
2026

- Decomposed a local AI chat app into containerised microservices (gateway, model, embedding, vector store, job queue, scheduler) orchestrated with Docker Compose.
- Placed an Nginx load balancer in front of 1-3 FastAPI gateway replicas, keeping one stable API port while replicas scale behind it.
- Built a lightweight autoscaler that polls queue depth every 5s and scales gateway replicas through Docker (<2 pending $\rightarrow 1$, $\geq 2 \rightarrow 2$, $\geq 5 \rightarrow 3$), keeping the Redis-backed queue a single writer.
- Routed inference to the least-busy healthy node, with background health checks every 15s removing failed nodes from rotation.
- Added retrieval with Ollama embeddings and ChromaDB, and full observability with Prometheus metrics and pre-provisioned Grafana dashboards for queue depth, active requests and replica count.

NestMate | Roommate Finder & Household Hub

[🔗 Live](#) | [📄 Client](#) | [📄 Server](#)

Full-Stack Developer — Next.js, TypeScript, Express.js, MongoDB, Firebase

May 2026

- Developed a roommate discovery platform with verified listings, favourites and direct host messaging with conversation tracking.
- Engineered a Household Hub for shared living: join-code access, member tracking and room-based organisation.
- Built an automated meal ledger and expense tracker with fair-share splitting for rent, groceries and monthly dues.
- Added real-time room chat with Firebase, and secured auth and persistence with Firebase Auth and MongoDB.

EDUCATION

Jagannath University	Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering · CGPA 3.25 / 4.00	2021 – Present
Relevant coursework: Algorithms, Database Systems, Software Engineering, System Design	
Birshrestho Munsir Abdur Rouf Public College	Dhaka, Bangladesh
Higher Secondary Certificate (Science) · GPA 5.00 / 5.00	2019 – 2020

ACHIEVEMENTS & INTERESTS

- Competitive Programming: Active problem solver on LeetCode and Codeforces, focused on efficient algorithm design.
- Open Source: Contributed to GitHub repositories, focusing on documentation improvements and bug fixes.
- Soft Skills: Technical writing, team collaboration in Agile environments, problem solving.
- Languages: English (fluent), Bangla (native).